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Exploring the Role of Physical Activity in Individuals with Comorbid Cancer and Dementia: A Scoping Review

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Physical activity · Exercise · Cancer · Dementia · Comorbid cancer and dementia

Abstract
Background: Comorbid cancer and dementia, which share common risk factors and significantly burden the healthcare system, affect a growing number of individuals, especially the ageing population. As both conditions place a substantial burden on healthcare systems and may be underdiagnosed, there is an urgent need to explore effective management strategies, including the potential benefits of physical activity, which has shown promise in mitigating cognitive decline and improving physical function in both cancer and dementia populations. This scoping review aimed to explore the current knowledge of physical activity for individuals with comorbid cancer and dementia, identifying gaps in understanding and highlighting the need for future research in this area. ***Summary:*** This scoping review followed the 5-stage framework outlined by Arksey and O'Malley, with a focus on identifying the effects of physical activity on individuals with comorbid cancer and dementia. The review involved a comprehensive search across multiple databases, selecting relevant studies based on predefined criteria, and summarizing key findings to highlight research gaps and inform future studies. Out of 263 records identified from multiple databases, none were retained for full-text screening due to exclusions based on review articles, non-human participants, lack of comorbid cancer-dementia, and absence of a physical activity/exercise component. ***Key Messages:*** There is a significant gap in research on physical activity in individuals with comorbid cancer and dementia. Future studies are essential to explore the impact of exercise on the development and outcomes of these conditions, which could improve preventative strategies and care pathways for this growing population.

Introduction

Cancer and dementia are non-communicable diseases, which share risk factors, including age [1, 2], cardiovascular health [3], and lifestyle factors [4, 5]. The term "comorbid cancer and dementia" refers to individuals living with both conditions simultaneously [6]. It is estimated that 1 in 13 adults over the age of 75 years are reportedly living with comorbid cancer and dementia [7]. Due to the complex nature of these diseases and the typical care pathways, these conditions can often be underdiagnosed, suggesting the actual prevalence may be higher than currently reported [8]. Studying comorbid

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cancer and dementia is essential due to the complex clinical challenges it presents. Individuals with both conditions often have worse overall survival [9], higher healthcare system utilization [7], and reduced quality of life [9, 10]. For example, they require 9% and 37% more general practice appointments compared to those with cancer alone and dementia alone, respectively [7]. Comorbid patients often experience delayed diagnoses [9, 10], undergo fewer treatments [9], and experience communication difficulties during medical appointments, reducing their capacity to make informed decisions about cancer treatments [9]. Furthermore, dementia increases susceptibility to cancer treatment side effects, reducing treatment tolerance, leading to worse cancer outcomes [9]. In nursing home settings, those with comorbid cancer and dementia experience more neuropsychiatric symptoms, including sleep disturbances and agitation, than those with cancer alone ($p < 0.05$); and they receive more analgesics compared with those with dementia alone ($p < 0.05$) [11]. Understanding these unique challenges is critical for developing targeted interventions and ensuring healthcare systems can provide integrated, person-centred care. With an ageing population contributing to a growing number of comorbid cases, there is an urgent need for tailored prevention and management strategies [7].

Promisingly, physical activity has demonstrated benefits for those living with dementia, with higher activity levels associating with increased life expectancy and quality of life [12]. A systematic review of exercise interventions found that supervised multi-modal exercise ~60 min/day, 2–3 days/week, leads to improvements in strength, balance, mobility, and endurance [13]. Furthermore, a systematic review of moderate intensity aerobic exercise interventions reduced cognitive decline and behavioural issues in dementia patients, with the most pronounced effect on lessening working memory decline (SMD 0.28, 95% CI: 0.04–0.52, $I^2 = 40\%$) [14]. Similarly, physical activity is highly beneficial for people with cancer, with higher levels of post-diagnosis physical activity associating with a 45% and 37% reduction in cancer recurrence and mortality, respectively [15]. A meta-analysis showed strong evidence for exercise interventions to improve physical function, fitness, muscular strength, and quality of life in those living with cancer [16]. Additionally, common side effects of cancer treatments, including cachexia [17], peripheral neuropathy [18], cardiotoxicity [19], depression [20], and fatigue [21] can be reduced with exercise interventions both during and after cancer treatments.

Given the established benefits of exercise for both cancer and dementia, it is reasonable to hypothesize that physical activity could also benefit individuals with comorbid cancer and dementia [15–18]. However, the clinical, functional, and cognitive complexities unique to this population – such as impaired memory, executive dysfunction, reduced dysfunction, and fragmented care pathways [6, 7, 9–11] – mean that these benefits cannot be assumed based on evidence from single cohorts. There is a critical need to explore safety, feasibility, and effectiveness of exercise interventions tailored to the needs of comorbid cancer and dementia. For example, those with cognitive impairment may require modified delivery methods, enhanced supervision, or caregiver involvement [22]. Similarly, cancer-related fatigue or ongoing treatment side effects may necessitate lower-intensity or more flexible exercise prescriptions [23]. As those with comorbid cancer and dementia have accelerated functional decline, greater symptom burden, worse treatment options and outcomes [9], PA/exercise could offer an adjunct to usual care, potentially improving health and quality of life. Targeted research will help identify appropriate exercise modalities, intensities, and supervision, ensuring interventions are safe and effective. This is essential for developing integrated, person-centred strategies to support health, independence, and wellbeing. This scoping review endeavours to explore current knowledge of physical activity/exercise for people with comorbid cancer and dementia, to determine our current understanding of the topic and identify any gaps in knowledge to ascertain the requirement for future research in this field.

Methods

This scoping review was conducted following the 5-stage framework outlined by Arksey and O'Malley [24]. Scoping reviews utilize narrative analytical methods and are generally not intended to evaluate the quality of individual studies [24]. As specified in the PRISMA extension for Scoping Reviews (PRISMA-ScR) [25], certain PRISMA items, such as assessing risk of bias across studies and performing additional analyses, are not applicable to scoping reviews. Consequently, these elements were not included in this review. The PRISMA-ScR checklist [25] is provided in online supplementary Appendix 1 (for all online suppl. material, see https://doi.org/10.1159/000547553).

Stage 1: Identify the Research Question

This review is guided by the following research question: "What is known about the effects of physical activity/exercise on individuals living with comorbid cancer and dementia?" Three key concepts were identified from the review question: "cancer" "dementia," and "physical activity/exercise."

Stage 2: Identifying Relevant Studies

This scoping review incorporated manuscripts that met the following inclusion criteria:

- Published in English in peer-reviewed academic journals
- Designs including primary research studies (cross-sectional or longitudinal intervention designs, randomized controlled trials, controlled trials, and uncontrolled trials)
- Target population: individuals with cancer and dementia (comorbid), of any age, in any setting
- Must include physical activity (defined as any bodily movement produced by skeletal muscles that results in energy expenditure) or exercise (defined as a subset of physical activity that is planned, structured, and repetitive and has as a final or an intermediate objective the improvement or maintenance of physical fitness).
- Publication year: no limits

Studies were excluded based on the following exclusion criteria (screening by title, abstract, and/or full text):

- Review articles;
- non-human subjects;
- no physical activity/exercise component;
- target population: other than comorbid cancer-dementia (except for healthy controls as a comparison group).

Search Strategy and Databases

Five electronic databases were searched during January 2025: MEDLINE, PubMed, Google Scholar, Scopus, and Cochrane Library. The search strategies are outlined in Table 1 and the PRISMA flow diagram ([25], Fig. 1) summarizes the search strategy and selection process.

Stage 3: Study Selection

All records identified through searching the three databases were exported to Zotero with abstracts and references. After duplicates were manually removed by the first author (M.M.), two researchers (M.M. and N.S.-H.) screened the remaining 260 record titles and abstracts. Both authors agreed that there were no records that met the pre-defined inclusion criteria.

Stage 4: Charting the Data

The authors aimed to excerpt the following key information from each included record: author(s), year of publication, study aim, methodological details, participant characteristics, interventions, outcome measures, and main findings.

Stage 5: Collating, Summarizing, and Reporting

The authors planned on using the "descriptive-analytical" method [25] to produce key summaries for each outcome to address the research question and highlight existing research gaps for future exploration.

Results

From the initial 263 records identified through searches in MEDLINE, PubMed, Scopus, Cochrane Library, and Google Scholar, 0 were retained for full-text screening. The records were excluded for reviews, non-human participants, not comorbid cancer-dementia, and did not include a physical activity/exercise component (shown in Fig. 1). The resultant zero records meant that the intended stage 4 and 5 could not be conducted.

Discussion

The lack of research investigating physical activity/exercise in populations with comorbid cancer and dementia is striking. Upon finding zero records when following the 5-stage PRISMA guidelines for scoping reviews [24, 25], the current manuscript could have been omitted from submission to publication. However, it is important that this finding is known to highlight the need for exploration in this area.

There are key areas, which should be addressed to fill the research gap within the comorbid cancer and dementia field. One is the pressing need to determine more accurately the prevalence of the conditions comorbidly as currently, 7.5% adults aged over 75 years are reportedly living with comorbid cancer and dementia [7]. However, this is likely to be a gross underestimation of the true problem due to the difficulty of obtaining a secondary diagnosis within complex conditions such as cancer and dementia [8]. Therefore, future studies should prioritize the development of robust methodologies to accurately identify and quantify the prevalence of the conditions. This should include the integration of large-scale, population-based cohorts with linkage to electronic health records and registries to enhance diagnostic accuracy [26].

As the elderly population continues to grow, there is a pressing need to find preventative measures against comorbid cancer and dementia [7]. There is strong evidence that physical activity has an inverse dose-response relationship with cancer development in non-dementia populations [27]. A recent meta-analysis of >30 million participants found a relative risk of 0.93 (95% CI: 0.91–0.95),

Table 1. Search strategy for PubMed, MEDLINE, Scopus, Cochrane Library, and Google Scholar

Database	Construct	Search terms (exact Boolean string)
PubMed	Cancer	(“cancer s” [All Fields] OR “cancerated” [All Fields] OR “canceration” [All Fields] OR “cancerization” [All Fields] OR “cancerized” [All Fields] OR “cancerous” [All Fields] OR “neoplasm” [All Fields] OR “tumor” [All Fields] OR “tumour” [All Fields] OR “malignant” [All Fields] OR “carcinoma” [All Fields])
PubMed	Dementia	(“dementia” [MeSH Terms] OR “dementia” [All Fields] OR “dementias” [All Fields] OR “Alzheimer disease” [MeSH Terms] OR “Alzheimer’s disease” [All Fields]
PubMed	Exercise/physical activity	(“exercise” [MeSH Terms] OR “physical activity” [All Fields] OR “physical activities” [All Fields] OR “aerobic exercise” [All Fields] OR “aerobic training” [All Fields] OR “resistance training” [All Fields] OR “strength training” [All Fields] OR “endurance training” [All Fields] OR “activity level” [All Fields] OR “exercises” [All Fields] OR “exercising” [All Fields])
MEDLINE	Cancer	(cancer* OR cancers* OR cancerated* OR canceration* OR cancerization* OR cancerized* OR cancerous* OR neoplasm* OR tumor* OR tumour* OR malignant* OR carcinoma*).mp.
MEDLINE	Dementia	(dementia* OR dementias* OR Alzheimer disease* OR Alzheimer’s disease*).mp.
MEDLINE	Exercise/physical activity	(“exercise” OR “physical activity” OR “physical activities” OR “aerobic exercise” OR “aerobic training” OR “resistance training” OR “strength training” OR “endurance training” OR “activity level” OR “exercises” OR “exercising”).mp.
Scopus	Cancer	TITLE-ABS-KEY(“cancer” OR “cancers” OR “cancerated” OR “canceration” OR “cancerization” OR “cancerized” OR “cancerous” OR “neoplasms” OR “tumor” OR “tumour” OR “malignant” OR “carcinoma”)
Scopus	Dementia	TITLE-ABS-KEY(“dementia” OR “dementias” OR “Alzheimer disease” OR “Alzheimer’s disease”)
Scopus	Exercise/physical activity	TITLE-ABS-KEY(“exercise” OR “physical activity” OR “physical activities” OR “aerobic exercise” OR “aerobic training” OR “resistance training” OR “strength training” OR “endurance training” OR “activity level” OR “exercises” OR “exercising”)
Cochrane Library	Cancer	(“cancer” OR “cancers” OR “cancerous” OR “cancerated” OR “canceration” OR “cancerization” OR “cancerized” OR “neoplasms” OR “tumor” OR “tumour” OR “malignant” OR “carcinoma”)
Cochrane Library	Dementia	(“dementia” OR “dementias” OR “Alzheimer disease” OR “Alzheimer’s disease”)
Cochrane Library	Exercise/physical activity	(“exercise” OR “physical activity” OR “physical activities” OR “aerobic exercise” OR “aerobic training” OR “resistance training” OR “strength training” OR “endurance training” OR “activity level” OR “exercises” OR “exercising”)
Google Scholar	Cancer	“cancer” OR “cancers” OR “cancerous” OR “cancerated” OR “canceration” OR “cancerization” OR “cancerized” OR “neoplasms” OR “tumor” OR “tumour” OR “malignant” OR “carcinoma”
Google Scholar	Dementia	“dementia” OR “dementias” OR “Alzheimer disease” OR ”Alzheimer’s disease”
Google Scholar	Exercise/physical activity	“exercise” OR “physical activity” OR “physical activities” OR “aerobic exercise” OR “aerobic training” OR “resistance training” OR “strength training” OR “endurance training” OR “activity level” OR “exercises” OR “exercising”

0.88 (95% CI: 0.85–0.92), and 0.8 (0.95% CI: 0.81–0.89) for total cancer incidence for 4.375 mMET-hours/week, 8.75 mMET-hours/week, and 17.5 mMET-hours/week, respectively [27]. Similarly, there is strong evidence for the prevention of dementia in non-cancer populations [28–30]. A meta-analysis showed that those in the highest physical activity groups had a relative risk of 0.80 (95% CI: 0.77–0.84), 0.86 (95% CI: 0.80–0.93), and 0.79 (95% CI: 0.66–0.95) for all-cause dementia, Alzheimer’s disease, and vascular dementia, respectively, when compared to those in the lowest physical activity group [30]. Since physical activity has a potential role to play in the prevention of

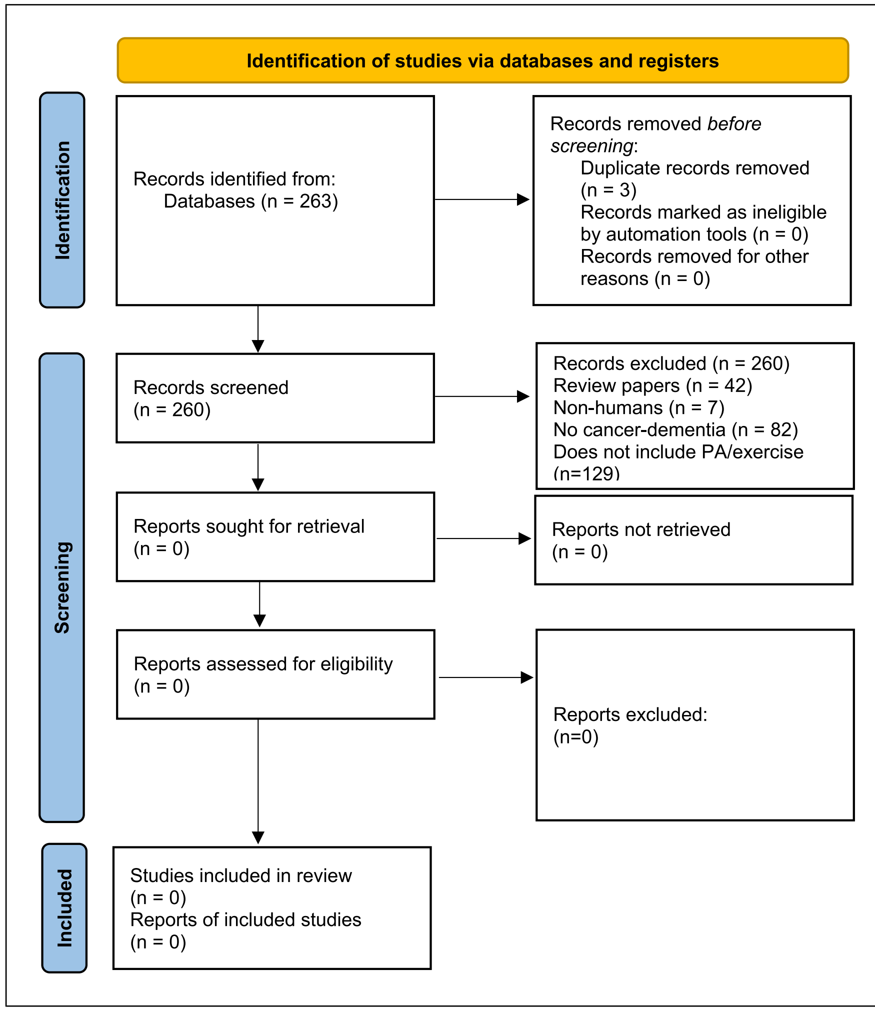

Fig. 1. PRISMA flow diagram for the scoping review process. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

single morbidity cancer [27] and dementia [28, 30], it is plausible that physical activity may associate with the risk of comorbid cancer and dementia. However, physiological complexity of the diseases and their treatments may attenuate potential benefits of physical activity in the development of the comorbid condition [6, 7]. As this scoping review found a complete absence of research into this area, it is pertinent to ascertain foundational evidence. Hence, future research should determine whether physical activity associates with the development of comorbid cancer and dementia: whether that be the development of dementia within cancer populations or vice versa. Utilizing longitudinal prospective [31] or retrospective cohort studies [32], cross-sectional studies [33], or nested case-controls within a cohort [34] will help to elucidate associations.

Furthermore, this scoping review had planned on assessing the effects of physical activity/exercise on physical and psychological outcomes within people living with comorbid cancer and dementia. There is a wealth of evidence for the benefits of physical activity/exercise in the adjunctive care of those living with cancer [16], as well as those living with dementia [13, 14, 35]. For people with a diagnosis of cancer, physical activity has reduced mortality risk in a dose-dependent manner, with a relative risk of 0.9 (95% CI: 0.88–0.93), 0.85 (95% CI: 0.81–0.89), and 0.82 (95% CI: 0.78–0.86) for 4.375 mMET-hours/week, 8.75 mMET-hours/week, and 17.5 mMET-hours/week, respectively [27]. There are well-documented benefits of physical activity/exercise for people living with cancer, with a systematic review of 4,519 participants showing that exercise interventions improved quality of life (effect size: 0.15, 95% CI: 0.10, 0.20) and physical function (effect size: 0.18, 95% CI: 0.13, 0.23) when compared to the usual care control groups [16]. Likewise, physical activity has reportedly reduced dementia-specific mortality risk in older adults, with a light physical activity of <3 h per week associating with a hazard ratio of 0.74 (95% CI: of 0.62–0.88) and of >3 h per week, with a hazard ratio of 0.61 (95% CI: 0.51–0.73) [36]. Participation in physical activity/exercise has been shown to benefit physical and psychological well-being of those living with dementia [13, 14, 35]. A meta-analysis reported that exercise interventions improved sit-to-stand performance by 2.1 repetitions (95% CI: 0.3–3.9), step length by 5 cm (95% CI: 2–8), balance by 3.6 points (95% CI: 0.3–0.7) on the Berg Balance Scale, reaching distance by 3.9 cm (95% CI: 2.2–5.5), walking speed by 0.13 m/s (95% CI: 0.03–0.24), distance covered in the 6-min walk test by 50 m (95% CI: 18–81) [13]. This highlights the meaningful physical impact of exercise participation in those with dementia. Additionally, exercise interventions have been shown to attenuate cognitive decline in those living with dementia, with a meta-analysis showing that exercise interventions improved global cognition (SMD: 0.45, 95% CI: 0.15–0.76), working memory (SMD: 0.28, 95% CI: 0.04–0.52), and behavioural problems (SMD: 0.36, 95% CI: 0.07–0.64), when compared to the no exercise control group [14]. Since there is evidence for benefits of exercise on physical and mental well-being in both dementia populations and cancer populations, it would be intuitive to explore the potential benefits of physical activity/exercise on comorbid cancer and dementia populations. However, the current scoping review found zero research into this area. This is highly concerning as there is a growing population of people living with the comorbid conditions.

The reason for the lack of studies is likely multifactorial. There are difficulties in researching the population due to underdiagnosis [8] and ethical concerns around capacity to consent or adherence to protocols due to cognitive impairments [9, 10]. High mortality, poor mobility, or worsening disease contribute to high attrition rates in longitudinal studies [10]. Additionally, historical silos between cancer and dementia research – due to disciplinary separation and distinct clinical pathways – have limited integrated approaches [37]. Consequently, despite rising prevalence, comorbid cancer and dementia populations remain under-researched [9, 10].

Future exploration into avenues such as physical activity/exercise for comorbid cancer and dementia is pertinent to improve knowledge, which may lead to implementation into the improvement of future care pathways. Research should involve liaising with patient and public involvement groups, patients, caregivers, and medical professionals, to design suitable interventions, which tackle complex needs of the population [38]. This will lead to safety and feasibility pilot trials, followed by randomized controlled trials to assess efficacy of exercise/physical activity on physical and psychological outcomes, utilizing mixed methods for in-depth assessment [39] of the optimal exercise for integration into survivorship or memory care programmes. It is worth noting that, despite the well-established benefits of physical activity/exercise in those living with cancer and dementia separately, both populations are less active than their healthy age-matched counterparts [23, 40]. There are specific barriers to physical activity participation in these populations, including fatigue, lacking motivation, and time limitations for those with cancer [23] and physical and mental limitations and difficulties with guidance and organization by caregivers in those with dementia [22]. Hence, it is essential that future research also determines the specific barriers to activity in comorbid cancer and dementia populations to support optimal implementation and adherence to future exercise interventions.

Conclusion

The complete lack of research on physical activity and exercise in individuals with comorbid cancer and dementia underscores a significant gap in the literature. Given the growing elderly population and the potential benefits of physical activity in both cancer and dementia populations, it is crucial to prioritize future studies exploring the impact of exercise on the development, physical, and psychological outcomes of comorbid cancer and dementia. Addressing this gap will not only enhance understanding but also inform preventative strategies and improve care pathways for this vulnerable population.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

M.M. conceived the scoping review. M.M and N.S.-H. contributed to screening records. M.M. and N.S.-H. drafted the full manuscript for submission.

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